

SkyLink International Airport

Business Analyst Portfolio Project 1 — Supporting Documents

Business Requirements Document

Section 1

BUSINESS REQUIREMENTS DOCUMENT

Airline Delay & Resource Optimization System

Client: SkyLink International Airport

Prepared for: VP, Revenue & Network Operations

Prepared by: Taruna Guliya, Business Analyst

Date: 24 June 2026

Status: Final v1.0

1. Document Control

Field	Detail
Document Title	Business Requirements Document — Airline Delay & Resource Optimization System
Client	SkyLink International Airport
Version	1.0
Author	Taruna Guliya, Business Analyst
Date Created	24 June 2026
Status	Final
Approvers	VP, Revenue & Network Operations; Director of Ground Operations

2. Business Objective

Providing the Revenue & Network Operations team of SkyLink International Airport with analysed data to quantify where flight delays, gate and counter congestion, and passenger load imbalances affected punctuality and operational efficiency during the May 2024 disruption episode — and to use those patterns to anticipate and prepare for the current recurrence of similar operational issues.

3. Background / Problem Statement

In May 2024, SkyLink's Network Operations experienced a short episodes of punctuality complaints and gate congestion during evening departures. At the time, no consolidated analysis linked flight schedule data, delay timings, gate/counter activity, and passenger load into a single operational view, so the root cause was never identified. The disruption eased after several weeks and operations stabilised without a formal investigation taking place.

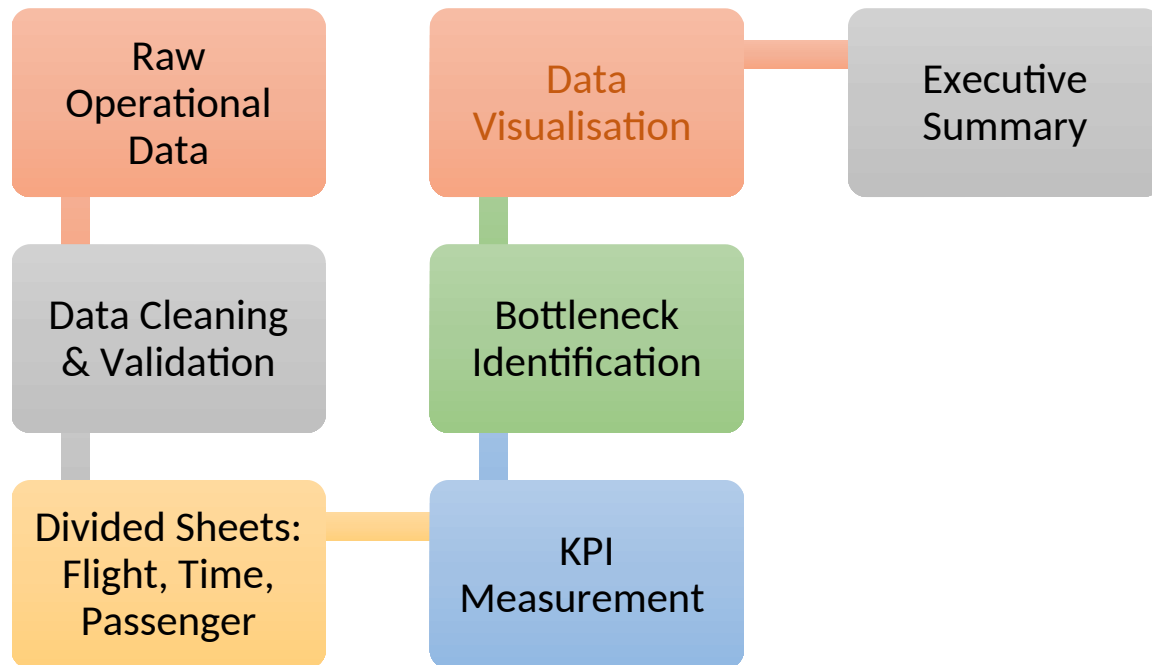
Similar punctuality and congestion symptoms have now resurfaced. Because the underlying cause of the May 2024 episode was never established, the Director of Ground Operations has commissioned this analysis to revisit the May 2024 operational data, identify the patterns and bottlenecks that characterised that episode, and use those findings to anticipate and prepare for the current recurrence ahead of decisions on gate reallocation, counter staffing, and carrier-level performance review.

4. Stakeholders

Role	Name / Title	Interest
Sponsor	VP, Revenue & Network Operations	Owns the budget decision on resourcing changes; needs evidence the problem is real and sized correctly
Business Owner	Director, Ground	Responsible for gate, stand, and counter

	Operations	allocation; primary user of the recommendations
End Users	Revenue & Ops Analysts	Will use the dashboard/pivot outputs monthly to track punctuality and congestion KPIs
Business Analyst	Taruna Guliya	Elicits requirements, defines scope, performs analysis, and reports findings

5. Process Flow



6. Scope

6.1 In Scope

- Analysis of flight-level delay data (arrival and departure) by route, carrier, and time of day
- Gate, stand, and check-in counter utilisation and congestion analysis
- Passenger load and occupancy analysis by flight type and hour of day
- Identification of peak congestion windows and their relationship to delay accumulation
- Data quality assessment of the source dataset, with issues flagged for stakeholder awareness
- Summary KPIs and recommendations delivered via spreadsheet pivot analysis and an executive summary

6.2 Out of Scope

- Crew scheduling
- Weather-related delay prediction or modelling
- Passenger compensation policy
- Real-time/live operational dashboards (this analysis uses a historical extract, not a live feed)
- Changes to airline contracts, slot allocation agreements, or airport infrastructure

7. Functional Requirements

Requirements are grouped under four business goals, consistent with how the analysis was structured and delivered.

Goal 1 — Analyse operational delays and identify major bottlenecks affecting punctuality

ID	Requirement	Maps to (Excel)
REQ-01	Classify each flight into a delay category (Early, On Time, Minor, Major, Severe) to distinguish normal variation from critical disruption.	Flights by Delay Category
REQ-02	Compare average arrival delay against average departure delay to identify whether disruption originates pre- or post-departure.	Delay Arrival vs Delay Departure

Goal 2 — Evaluate airport resource utilisation (gates, stands, counters)

ID	Requirement	Maps to (Excel)
REQ-03	Rank gates by flight volume to identify the highest-traffic / most congested gates.	Gate Utilisation Rate pivot (Top 10 gates)
REQ-04	Rank check-in counters by usage volume to identify over- and under-utilised counters.	Counter Usage pivot (Top 10 counters)

Goal 3 — Monitor passenger flow and aircraft load patterns

ID	Requirement	Maps to (Excel)
REQ-05	Segment flights into Low / Medium / High occupancy bands to assess seat utilisation efficiency.	Occupancy Rate Distribution pivot
REQ-06	Compare total passenger volume between arrival and departure flights.	Passenger Volume by Flight Type pivot
REQ-07	Identify the flights contributing the highest passenger volumes for prioritisation.	Top 10 Flights by Passenger Count pivot

Goal 4 — Identify peak operational hours contributing to congestion and delay

ID	Requirement	Maps to (Excel)
REQ-08	Profile flight volume by hour of day to locate the network's peak traffic window.	Flights by Departure Hour pivot
REQ-09	Profile average delay by hour of day to test whether delay correlates with traffic volume or occurs independently of it.	Average Delay by Departure Hour pivot
REQ-10	Profile passenger volume by hour of day to identify the passenger demand peak.	Passenger Volume by Hour pivot
REQ-11	Cross-tabulate gate activity by hour to produce a congestion heatmap identifying which gates are busiest and when.	Gate Congestion Heatmap pivot

8. Non-Functional Requirements

- The analysis must be refreshable monthly as new operational data becomes available, without rebuilding the underlying pivot structure
- Outputs must be interpretable by non-technical stakeholders — every pivot must be paired with a plain-language key insight, not raw numbers alone
- Findings intended for senior leadership must be summarised on no more than one to two pages (executive summary)
- All data transformations must be traceable from raw source through to final reporting layer, to support audit and quality review

9. Data Sources

Airline operational dataset obtained via third-party extract, covering flight schedules, delay timings, gate/stand activity, counter usage, and passenger load records.

Stage	Detail
Raw extract	124 columns, 5,921 rows
Cleaned dataset	86 columns, 5,920 rows

10. Assumptions & Constraints

- Analysis is based on a fixed historical extract; it does not reflect schedule changes made after the extract date
- Counter utilisation findings are constrained by sparse data — 90 to 95% of counter allocation fields were empty in the source file, limiting confidence in counter-level conclusions

- A small number of passenger records show seat occupancy above 100% of aircraft capacity; these rows were flagged rather than silently corrected, and overstate passenger load analysis until resolved
- This analysis is based on the May 2024 operational extract — the period of the original disruption episode — and is used as a historical aid for the current recurrence. Findings should be validated against present-day schedules, fleet, and resourcing before being applied to current decisions, as conditions may have changed since 2024
- 3 records (0.05%) carry flight dates from 2022–2023, inconsistent with the rest of the extract (May 2024). These were retained in row counts but should be investigated as likely data entry or extraction errors before being used in any date-based filtering logic.

11. Success Metrics / KPIs

KPI	Result
On-time performance	96% of flights departed on time; 3.9% experienced major/severe delay
Average delay	2.3 minutes average; international departures higher than domestic
Peak congestion window	3 PM–2 AM peak traffic; 4 AM shows highest delay despite lower volume
Gate & stand concentration	Top 10 gates handle 379 flights; top 7 stands handle 416 flights
Aircraft occupancy	87.1% average occupancy; over 80% of flights in Medium–High bands
Passenger flow	Departures carry more passengers than arrivals (178,790 vs 155,974); peak 214.8 pax/flight at 8 PM

12. Sign-Off

By signing below, stakeholders confirm that the business objective, scope, and requirements documented above accurately reflect the agreed analysis to be delivered.

Name / Role	Signature	Date
VP, Revenue & Network Operations (Sponsor)		
Director, Ground Operations (Business Owner)		

Executive Summary

Section 2

Project-1

SkyLink International's Airline Delay & Resource Optimization

Main Business Problem:

SkyLink International Airport faced airline and airport operations delays, resource allocation inefficiencies, and operational bottlenecks that impact punctuality and passenger flow.

Goals:

1. Analyse operational delays and identify major bottlenecks affecting flight punctuality.
2. Evaluate airport resource utilization including gates, stands, and boarding operations.
3. Monitor passenger flow and aircraft load patterns to identify operational inefficiencies.
4. Identify peak operational hours contributing to airport congestion and delay accumulation.

Tool Used: Excel

Dataset Source:

Airline operational dataset obtained through a third-party source containing flight schedules, delay timings, gate activities, and airport operational records used for operational efficiency analysis.

Contained:

Columns- 124

Rows:5921

Final Product Contain:

Columns- 86

Rows:5920

Data Quality issues:

1. Passenger Sheet

Total Passengers

Passenger count exceeds aircraft capacity — seat fill above 100% which is physically impossible

Passenger load analysis overstated

2. Flight Sheet

Counter 1–6

SkyLink International Airport is a hypothetical airport made for the sake of this project. The data is collected from third party source and used for data analysis and business requirements and results.

Counter allocation 90–95% empty — data too sparse for reliable analysis.

Utilisation and analysis are unreliable

Key KPIs:

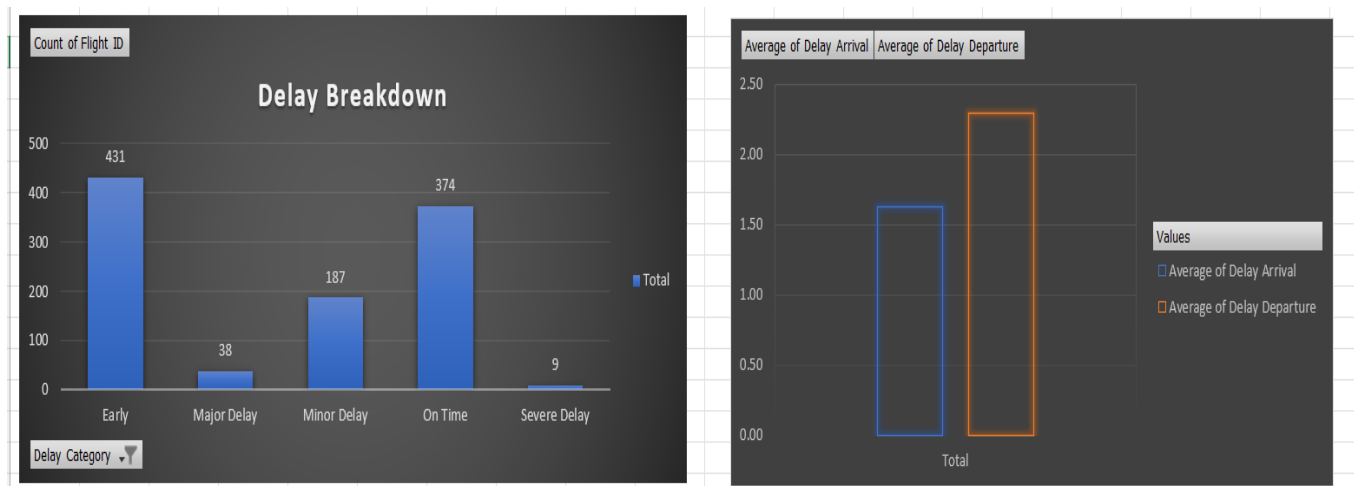
1. **On-time performance:** 96% of flights departed on time; severe delays affected only 3.9% of flights.
2. **Average delay:** Average delay was 2.3 minutes, with international departures experiencing higher delays than domestic flights.
3. **Peak congestion window:** Flight traffic peaked between 3 PM–2 AM, while the 4 AM slot showed the highest delays despite lower traffic.
4. **Gate & stand concentration:** The top 10 gates handle 379 flights and the top 7 stands handle 416 flights, with gates D34, D35, D40, and D42 repeatedly topping the busiest gate-hour combinations.
5. **Aircraft occupancy:** Average occupancy rate is 87.1%, with over 80% of flights in the Medium-to-High occupancy bands, reflecting strong seat utilization.
6. **Passenger flow:** Departure flights carry more passengers than arrivals (178,790 vs. 155,974 total), and passenger volume peaks between 5 PM and 10 PM, reaching 214.8 passengers per flight at 8 PM.

Recommendations:

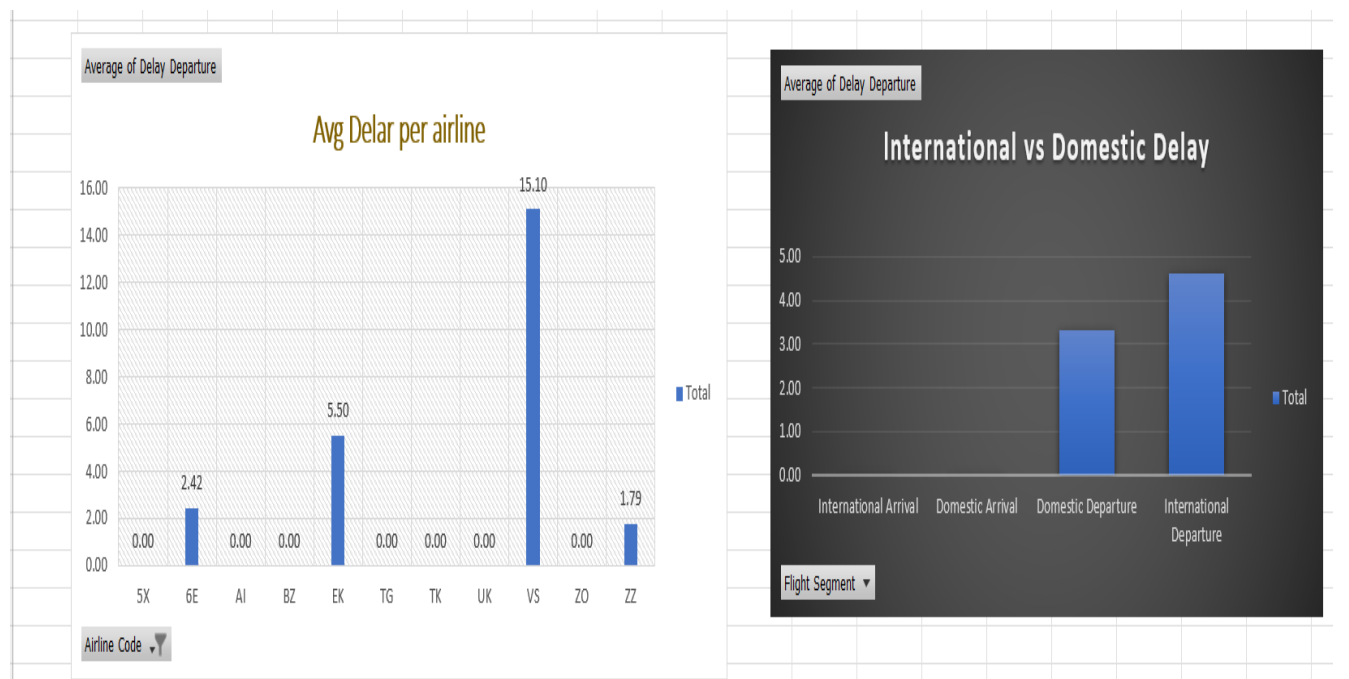
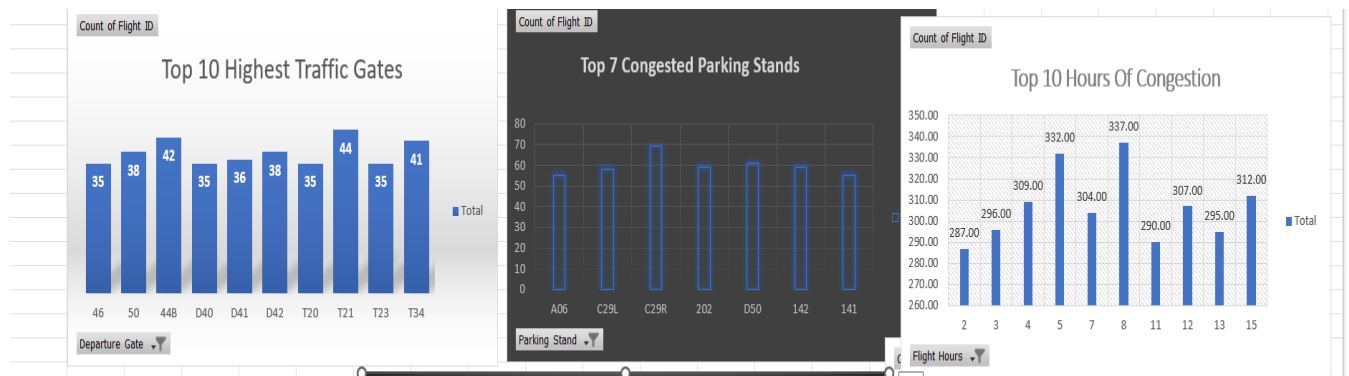
1. Prioritize the 4 AM departure slot for a turnaround-process review, since its delay rate is 3.5x the network average despite below-peak traffic volume, suggesting a fixable operational bottleneck rather than a capacity constraint.
2. Redistribute traffic from the busiest gates and stands (notably D34, D35, D40, D42, C29R, and 202) to underused alternatives to relieve congestion during the 3 PM–2 AM peak window.

3. Conduct a targeted operational review of VS, EK, 6E, and ZZ, since these carriers drive the highest average delays and offer the largest improvement opportunity for network-wide punctuality.
4. Focus on pre-departure activities investigation on international departures specifically, since they show notably higher delays than domestic departures and arrivals show comparatively minimal delay.
5. Scale staffing and resources for the 5 PM–10 PM passenger surge to match the departure-heavy, high-occupancy load already observed in this window.
6. Resolve the underlying data quality gaps (tow-date mismatches, sparse counter allocation, and pax-over-capacity rows, detailed below) before drawing turnaround-time or counter-utilization conclusions, since current coverage limits confidence in those specific metrics.

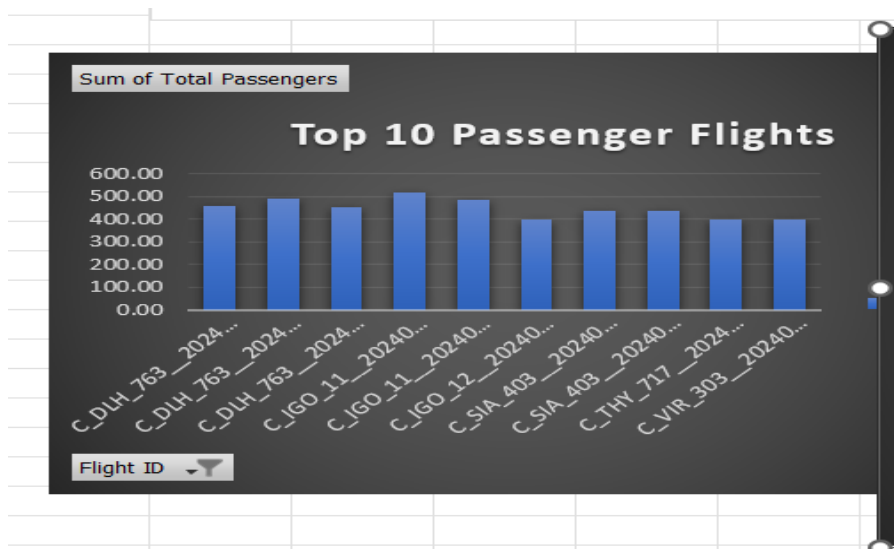
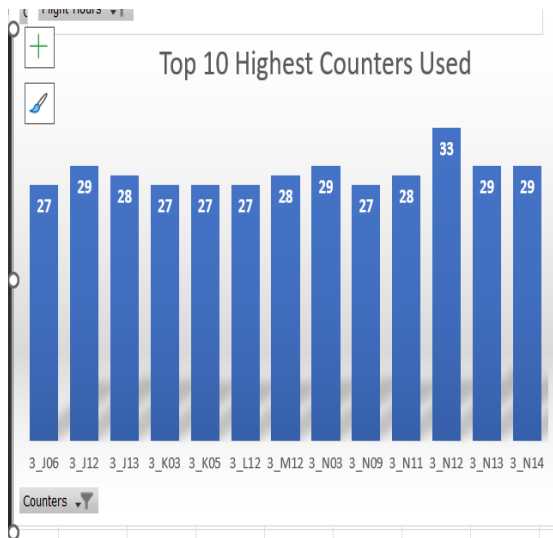
Dashboard Screenshots:



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